PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

LintasNiaga

Choose the best-value supplier with less hidden risk

Project Name	LintasNiaga
Team	404NotFounders
Prepared For	UMHackathon 2026
Target Domain	AI for Economic Empowerment & Decision Intelligence
Primary User	Malaysian plastics SME procurement executive / operations-commercial decision maker
Primary Product Scope	PP resin import quotation analysis, FOB only, Malaysia import context
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1. Project Overview

Project Name	LintasNiaga
Version	1.0 submission draft
Problem Statement	Malaysia has a sizable plastics manufacturing base, but many SME procurement teams still make raw-material purchasing decisions through fragmented supplier PDFs, spreadsheets, manual FX assumptions, and outdated freight or tariff baselines. This is especially risky for PP resin importers, where a supplier quote is not the true cost of purchase: the final landed cost is affected by currency movement, freight, import tariff, MOQ lock-up, supplier reliability, oil-linked logistics pressure, weather disruption, holidays, and market benchmark movement. For Malaysian plastics SMEs, a “cheap” quote can become expensive after hidden landed-cost drivers are included. At the same time, delaying a purchase can expose the company to FX and market volatility. The core problem is not simply quote extraction; it is procurement decision uncertainty.
Target Domain	AI for Economic Empowerment & Decision Intelligence; specifically cross-border procurement decision support for Malaysian SME importers.
Proposed Solution Summary	LintasNiaga converts supplier quotation PDFs into structured procurement inputs, refreshes market and risk context, calculates landed-cost scenarios, simulates p10/p50/p90 downside bands, and produces a traceable supplier, timing, and hedge recommendation.

In plain terms, LintasNiaga helps an SME answer: Which supplier should I choose, should I lock now or wait, and how much FX exposure should I hedge?

2. Background & Business Objective

Background of the Problem

The Malaysian plastics market is large enough for procurement inefficiency to matter. Mordor Intelligence estimates the Malaysian plastics market at USD 4.19 billion in 2026, with a forecast increase to USD 5.07 billion by 2031. This supports the view that plastics manufacturing is not a niche problem, but a meaningful industrial segment with recurring raw-material purchasing needs.

Within this sector, plastics manufacturers are largely SME-heavy. MPMA's member data indicates that its roughly 750 members represent about 60% of plastics manufacturers in Malaysia, while industry reporting has described the plastics manufacturing base as about 1,200 companies, with around 90% being SMEs.

For these firms, PP resin procurement is not a simple "pick the lowest unit price" decision. A quote must be normalized across supplier origin, currency, MOQ, lead time, incoterm, freight, tariff, supplier reliability, and market benchmark context. SMEs often lack the treasury, market-intelligence, and procurement-analytics capacity available to larger enterprises, so the same decision is frequently made using spreadsheets and intuition.

Importance of Solving This Issue

- **Margin protection:**
  A small percentage error in imported raw material cost can reduce SME margin materially.
- **Faster decision cycles:**
  Procurement teams need to act while quotations are still valid, not after slow manual spreadsheet comparison.
- **Better risk visibility:**
  The user should see downside exposure, not just a single average landed-cost number.
- **Explainability:**
  The final recommendation must be understandable by non-technical SME managers.

Strategic Fit / Impact

LintasNiaga fits the hackathon theme because it is a decision-intelligence system, not a generic chatbot. The system interprets structured and unstructured data, reasons over procurement trade-offs, recommends concrete actions, and explains decisions in plain language. It also connects to sponsor-relevant themes: SME enablement, cross-border commerce, FX/treasury risk, and GLM-powered reasoning.

3. Product Purpose

3.1. Main Goal of the System

The main goal is to help Malaysian SME importers make procurement decisions with facts and intelligence, not guesswork.

LintasNiaga should turn supplier quote PDFs plus market context into a clear procurement action: recommended supplier, lock-now-or-wait timing, and suggested hedge percentage.

3.2. Intended Users (Target Audience)

- Primary: procurement executive at a Malaysian plastics SME importing PP resin.
- Secondary: SME owner, finance manager, operations manager, or commercial decision maker who needs to approve raw-material purchases.
- Demo stakeholder: hackathon judges and mentors who need to see that the AI layer is genuinely involved in reasoning, not only UI narration.

3.3. User Goals

- Compare multiple supplier quotes quickly and fairly.
- Avoid misleadingly cheap quotes that hide freight, MOQ, reliability, or market-benchmark risk.
- Understand p10/p50/p90 landed-cost exposure over the procurement window.
- Decide whether to lock now, wait, or stage the order.
- Generate a bank-instruction style draft to communicate a hedge request.

4. System Functionalities

4.1. Description

The system operates as a backend-owned procurement analysis workflow. The frontend guides upload, review, and decision visualization, while FastAPI owns the actual extraction, validation, data refresh, landed-cost calculation, simulation, AI reasoning, traceability, and recommendation assembly.

The architecture is intentionally narrow: PP resin, FOB supplier quotes, Malaysia import context, and up to five supplier quotes per run. This narrow wedge keeps the demo credible and prevents the system from becoming a broad but shallow sourcing chatbot.

4.2. Key Functionalities

Functionality	What it does	Why it matters
Quote Upload	User uploads supplier quotation PDFs and enters quantity plus urgency.	Turns messy supplier documents into a structured workflow.
Quote Extraction	PDF text extraction runs first; GLM vision extraction is used as a fallback when text parsing is insufficient.	Keeps cost and latency lower while still handling difficult PDFs.
Review & Repair	User reviews and fixes supplier name, origin, incoterm, currency, unit price, MOQ, and lead time.	Prevents blind AI extraction errors from entering final analysis.
Validation Gate	Only valid or repaired quotes proceed. Unsupported incoterms, currencies, or origins are blocked.	Keeps comparison fair and honest.
Snapshot Refresh	FX, Brent oil, weather, macro, news, holidays, and PP resin benchmark context are refreshed or validated before analysis.	Avoids stale market assumptions.
Landed-Cost Engine	Computes material cost, freight, tariff, MOQ penalty, trust penalty, and total landed cost.	Gives a real procurement economics view, not just supplier unit price.
Monte Carlo Fan Chart	Generates p10/p50/p90 landed-cost bands using quote currency FX, USD/MYR freight conversion, and Brent-linked freight pressure.	Shows downside risk instead of one fragile average number.
PP Benchmark Check	Compares quote pricing against SunSirs PP resin benchmark and labels quote-vs-market risk.	Helps detect premium or suspiciously low quotes.
AI Recommendation	A thin LangGraph/GLM reasoning step consumes structured backend context and outputs bounded JSON.	Makes reasoning explainable while keeping math outside the model.
Hedge Replay	User adjusts hedge slider; backend recalculates risk bands using stable seeded shocks.	Shows how hedging narrows FX downside without rerolling the future.
Bank Instruction Draft	Backend drafts hedge instruction content; frontend renders a downloadable PDF.	Makes the recommendation action-oriented for bank/finance workflow storytelling.
Traceability	Langfuse traces extraction, recommendation reasoning, stream explanation, and related AI calls when configured.	Supports judge validation and debugging.

4.3. AI Model & Prompt Design

4.3.1. Model Selection

LintasNiaga uses a Z.AI GLM-family provider. In the current repository, the provider is configured through MODEL_API_KEY, MODEL_BASE_URL, and MODEL_NAME, with the implementation defaulting to an ilmu-glm-5.1 style model name.

The model is suitable because the product needs more than classification. It needs to read messy quote images when deterministic parsing fails, interpret structured cost outputs, reason over trade-offs, and produce a plain-language explanation that a procurement or finance user can defend.

The model is deliberately not the source of truth for numeric economics. Deterministic backend services calculate landed cost, ranking, FX/oil simulation, hedge replay, tariff, freight, and supplier penalties. The GLM reasons over those outputs and explains the decision.

4.3.2. Prompting Strategy

The prompting strategy is structured, multi-step, and schema-bounded rather than casual chat prompting.

- **Quote extraction prompt:** The model is asked to return strict JSON fields only, such as supplier name, origin, incoterm, unit price, currency, MOQ, lead time, and payment terms.
- **Recommendation prompt:** The model receives ranked supplier quotes, landed-cost breakdowns, FX simulation results, PP benchmark context, and market/risk context. It must return strict JSON.
- **Bounded decision rule:** The model may only confirm or swap supplier recommendations between rank #1 and rank #2, and only for specific reasons such as reliability, downside risk, MOQ lock-up, urgency mismatch, or disruption risk.
- **Plain-language explanation:** The model must explain reasons in SME-friendly language and connect them to visible fan chart, risk drivers, news, market snapshots, or benchmark signals.
- **Streaming analyst prompt:** A separate streaming explanation summarizes the recommendation in a few readable paragraphs for the result page.

4.3.3. Context & Input Handling

- **PDF handling:** The system accepts PDF quote uploads. The current quote extraction reads or renders the first two pages for extraction.
- **Upload limit:** The frontend workflow supports up to five supplier quotes per analysis run.
- **Mandatory user inputs:** At least one valid quote, required quantity in metric tons, urgency, and hedge preference are needed before analysis.
- **Scope filtering:** Quotes must fit the current wedge: FOB incoterm, supported currencies, and recognized China/Thailand/Indonesia origins.

- **Oversized or messy inputs:** If quote fields cannot be extracted confidently, the user is routed to review/repair. If the quote is out of scope, it is excluded instead of forced into the analysis.
- **AI context compression:** The model is not given raw full datasets. The backend builds a structured context summary containing ranked quotes, cost results, selected scenario, benchmark, and top risk signals.

4.3.4. Fallback & Failure Behavior

- **Extraction fallback:** PDF text extraction runs first. GLM vision extraction is used when deterministic parsing fails or produces an invalid quote.
- **Human repair path:** Users can correct extracted fields before analysis. This is the main human-in-the-loop safeguard.
- **Invalid quotes:** Fixable invalid quotes stay in the repair workflow. Out-of-scope quotes are excluded honestly.
- **No valid quotes:** Analysis is blocked and the user is asked to fix or upload new quotes.
- **Model output failure:** Recommendation reasoning requires valid JSON and a captured trace URL. If this fails, analysis fails visibly instead of fabricating a recommendation.
- **Bank draft fallback:** The bank-instruction draft has a deterministic fallback so the demo workflow can continue even if the wording model fails.
- **Snapshot failure:** Critical market context failure can block analysis. The product favors explainable failure over fake certainty.

5. User Stories & Use Cases

5.1. User Stories

User Story	Acceptance Outcome
As a procurement executive, I want to upload supplier quotation PDFs so that I do not need to manually retype every field.	The system extracts fields and shows them for review.
As a procurement executive, I want to correct extracted fields before analysis so that bad extraction does not produce a bad decision.	The review page allows field repair and revalidation.
As an SME decision maker, I want to compare quotes by landed cost, not just unit price, so that I avoid hidden cost traps.	The result ranks suppliers using material, freight, tariff, MOQ, and trust penalties.
As a finance-aware buyer, I want to see p10/p50/p90 risk bands so that I understand the downside before committing.	The results page shows a fan chart and risk-width metrics.
As a manager, I want to know why the winning supplier was chosen so that I can approve the decision confidently.	The AI analyst panel provides plain-language reasons and why-not explanations.
As a user with only one usable quote, I want an honest evaluation instead of fake comparison.	The system switches to single-quote mode with proceed/review/do-not-recommend labels.

As a finance user, I want to adjust the hedge percentage so that I can see how FX protection changes downside risk.	The hedge slider calls backend replay and updates the chart without rerolling shocks.

5.2. Use Cases (Main Interactions)

1. Multi-quote comparison: user uploads two to five quotes, reviews extracted fields, runs analysis, and receives ranked supplier recommendations.
2. Single-quote evaluation: user uploads one valid quote or only one quote remains valid; system evaluates whether to proceed, review carefully, or avoid.
3. Quote repair: user corrects missing or wrong fields before analysis.
4. Market-aware decision: system refreshes snapshots and compares quotes against FX, Brent, weather, macro, news, holidays, and SunSirs PP benchmark context.
5. Hedge replay: user moves the hedge slider to see how p90 downside changes.
6. Bank instruction draft: user downloads a structured hedge instruction PDF for a bank-style workflow demonstration.
7. Traceability check: judge or teammate opens Langfuse trace to confirm that GLM reasoning was actually invoked.

5.3. Use Case Diagram (Textual)

Step	User/System Action	Purpose
1	SME user uploads supplier quote PDFs	Start quote comparison using real supplier documents
2	System extracts quote fields	Convert unstructured PDFs into structured procurement inputs
3	User reviews and repairs fields	Prevent bad extraction from entering analysis
4	Backend runs cost + risk engine	Calculate landed cost, freight, tariff, MOQ, FX, and risk bands
5	GLM reasons over structured results	Produce plain-language recommendations and caveats
6	Dashboard shows recommendation	Display supplier choice, timing, hedge %, risk drivers, and fan chart

7	User adjusts hedge / generates bank PDF	Explore FX protection and prepare an execution draft

6. Features Included (Scope Definition)

- PDF quotation upload and structured quote state creation.
- Deterministic text extraction first, with GLM vision fallback for difficult PDFs.
- User review and repair of extracted supplier fields.
- Validation gate for FOB, supported currencies, recognized origins, and required quote fields.
- Quantity-based landed-cost calculation.
- Freight, tariff, MOQ, and supplier trust/reliability components.
- FX and Brent-driven Monte Carlo landed-cost fan chart.
- SunSirs PP resin benchmark fairness check.
- Risk drivers using weather, holidays, macro, news, oil, FX, freight, tariff, and PP benchmark context.
- Comparison mode and single-quote mode.
- Bounded GLM recommendation with LangGraph orchestration.
- Langfuse traceability for AI calls when credentials are configured.
- Hedge slider replay through the backend endpoint.
- Bank-instruction draft generation and frontend PDF download.
- Snapshot ingestion and latest snapshot endpoints for supported datasets.

7. Features Not Included (Scope Control)

- No direct trade execution or live WorldFirst transaction execution in the current version.
- No ERP integration or accounting-system sync.
- No full enterprise authentication/authorization layer.
- No production-grade persistent database store for active quote/run state; active state is currently memory-backed.
- No broad commodity coverage beyond the PP resin wedge.
- No non-FOB supplier comparison in the current validated scope.
- No fully automated supplier onboarding or long-term supplier relationship management.
- No guarantee that every external provider is always available; strict failures are surfaced where needed.
- No production legal/tariff guarantee; tariff/freight data must be officially verified before real commercial use.
- No frontend automated E2E test suite is asserted as complete in this PRD.

8. Assumptions & Constraints

8.1. Product Assumptions

- The initial user segment is Malaysian plastics SMEs importing PP resin.
- The primary decision is supplier selection, order timing, and FX hedge percentage.
- Quotes are expected to include supplier name, origin, incoterm, unit price, currency, MOQ, and lead time.
- FOB is the supported incoterm for comparable landed-cost analysis.
- Port Klang / Malaysia import context is the assumed destination context.
- Supported origin context currently focuses on China, Thailand, and Indonesia.

8.2. Technical Constraints

- Current stack is a two-app monorepo: Next.js frontend and FastAPI backend.
- Current storage is local file-backed snapshots/raw artifacts/uploads plus in-memory active quote/run state.
- The frontend is thin; business logic remains in FastAPI services.
- External data is normalized into snapshots before being used in analysis.
- Langfuse credentials are required for strict AI traceability in the latest backend path.
- Some analysis latency depends on external refresh success and provider response time.

8.3. Performance Constraints

- Quote extraction can be slower when GLM vision fallback is required.
- Analysis may wait for FX, Brent, weather, macro, news, resin, and holiday freshness checks.
- Monte Carlo simulation is intentionally bounded for hackathon responsiveness.
- The system prefers clear blocking failures over hidden stale-data certainty when critical context is unavailable.

8.4. LLM Cost Constraint

Exact token cost depends on the final model endpoint pricing and the final number of uploaded quotes. The current product design reduces inference cost by using deterministic extraction first, limiting quote extraction to early pages, summarizing context before recommendation, requiring strict JSON output, and keeping deterministic math outside the model.

Cost-control decision	Reason
Deterministic text extraction before GLM vision	Avoids model calls for clean PDFs.
Maximum five quotes per run	Bounds extraction and recommendation context size.
Structured backend context instead of raw datasets	Keeps prompt size manageable.
Strict JSON prompts	Reduces verbose model output and easier parsing failure detection.

AI only after deterministic computation	Prevents repeated model calls for math that code can do reliably.

8.5. User Input Constraint

The model does not autonomously decide from raw documents. The user must upload quotes, provide the required quantity, confirm or repair extracted fields, and trigger analysis. Human review is part of the product design, not an afterthought.

9. Risks & Questions Throughout Development

9.1. Risks

Risk	Mitigation / Current Position
External provider instability	Use normalized snapshots, freshness checks, clear failure messages, and local evidence where applicable.
Model hallucination or invalid output	Strict JSON schema, bounded top-2 supplier selection, deterministic fallback where appropriate, and Langfuse tracing.
Bad quote extraction	PDF text extraction first, GLM fallback, user repair page, and validation gate before analysis.
Overclaiming live market truth	PP benchmark is labeled as benchmark context, not exact supplier truth. Freight/tariff references require careful labeling.
In-memory state loss	Acceptable for hackathon demo, but must be replaced by durable persistence for production.
Scope limitation	Current version remains PP resin, FOB, Malaysia import context, up to five quotes.
Frontend QA gap	Backend tests exist; frontend manual end-to-end verification remains important before final demo.

9.2. Final Development Position

LintasNiaga is strongest when framed as a focused, evidence-first procurement decision system. Its core advantage is not that it automates every procurement task. Its advantage is that it combines quote extraction, deterministic cost economics, refreshed market context, bounded GLM reasoning, hedge simulation, and traceability into one decision workflow.

Appendix A. Implementation Grounding Summary

- Frontend: Next.js 16, React 19, TypeScript, Tailwind CSS, React Query, Recharts, React PDF, SSE consumption.
- Backend: FastAPI, Pydantic, Uvicorn, PyMuPDF, pandas/numpy, yfinance, OpenWeather, OpenDOSM, GNews, SunSirs/trafilatura, LangGraph, Langfuse.
- Data: local reference JSON, normalized snapshots, raw artifacts, uploaded PDFs, in-memory active run state.

- AI: GLM provider wrapper with Langfuse callbacks; thin LangGraph reasoning node; deterministic business logic remains outside AI.

Appendix B. Short Judge-Friendly Product Statement

LintasNiaga helps a Malaysian SME procurement user upload supplier quotes, normalize landed costs, understand downside risk, and receive a traceable recommendation on supplier, timing, and hedge ratio. The backend does the math; the GLM explains the decision using structured evidence. The user gets a clear action, not just another dashboard.